



AWS Application Integration Service

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integration services are a family of services that enable decoupled communication between applications.



AWS Application Integration

These services provide decoupling for microservices, distributed systems, and serverless applications.

AWS application integration services allow you to connect apps, without needing to write custom code to enable interoperability.

Decoupled applications can interoperate whilst being resilient to the failure or overload of any individual component.

The following services are involved with application integration:

Service	What it does	Example use cases

- [Amazon Simple Notification Service](#)
- [Amazon Simple Queue Service \(Amazon SQS\)](#)
- [Amazon Simple Workflow Service \(Amazon SWF\)](#)
- [Amazon MQ](#)
- [AWS Step Functions](#)

Simple Queue Service (SQS)	Messaging queue; store and forward patterns	Building distributed / decoupled applications
Simple Notification Service (SNS)	Set up, operate, and send notifications from the cloud	Send email notification when CloudWatch alarm is triggered
Step Functions	Out-of-the-box coordination of AWS service components with visual workflow	Order processing workflow
Simple Workflow Service (SWF)	Need to support external processes or specialized execution logic	Human-enabled workflows like an order fulfilment system or for procedural requests
Amazon MQ	Message broker service for Apache Active MQ and RabbitMQ	Need a message queue that supports industry standard APIs and protocols; migrate queues to AWS

Amazon Simple Notification Service

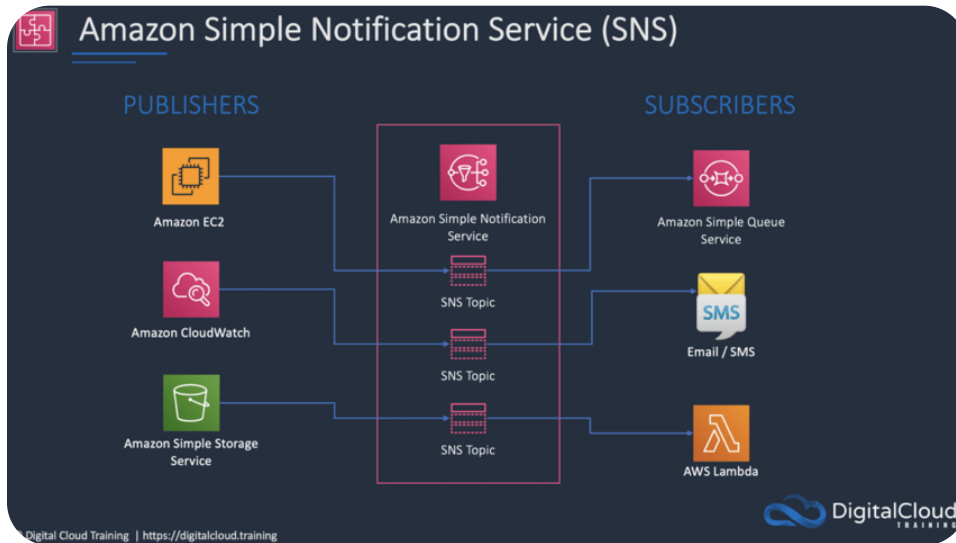
Amazon Simple Notification Service (Amazon SNS) is a web service that makes it easy to set up, operate, and send notifications from the cloud.

Amazon SNS is used for building and integrating loosely-coupled, distributed applications.

SNS provides instantaneous, push-based delivery (no polling).

SNS concepts:

- Topics – how you label and group different endpoints that you send messages to.
- Subscriptions – the endpoints that a topic sends messages to.
- Publishers – the person/alarm/event that gives SNS the message that needs to be sent.



SNS usage:

- Send automated or manual notifications.
- Send notification to email, mobile (SMS), SQS, and HTTP endpoints.
- Closely integrated with other AWS services such as CloudWatch so that alarms, events, and actions in your AWS account can trigger notifications.

Uses simple APIs and easy integration with applications.

Flexible message delivery is provided over multiple transport protocols.

Offered under an inexpensive, pay-as-you-go model with no up-front costs.

The web-based AWS Management Console offers the simplicity of a point-and-click interface.

Data type is JSON.

SNS supports a wide variety of needs including event notification, monitoring applications, workflow systems, time-sensitive information updates, mobile applications, and any other application that generates or consumes notifications.

SNS Subscribers:

- HTTP.
- HTTPS.
- Email.
- Email-JSON.
- SQS.
- Application.
- Lambda.

SNS supports notifications over multiple transport protocols:

- HTTP/HTTPS – subscribers specify a URL as part of the subscription registration.
- Email/Email-JSON – messages are sent to registered addresses as email (text-based or JSON-object).
- SQS – users can specify an SQS standard queue as the endpoint.
- SMS – messages are sent to registered phone numbers as SMS text messages.

Topic names are limited to 256 characters.

SNS supports CloudTrail auditing for authenticated calls.

SNS provides durable storage of all messages that it receives (across multiple AZs).

Amazon Simple Queue Service (Amazon SQS)

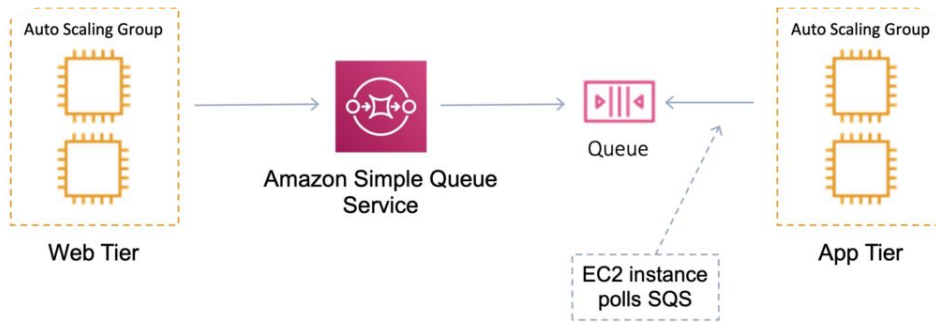


Amazon Simple Queue Service (SQS) is a distributed queue system.

Amazon SQS enables you to send, store, and receive messages between software components.

An Amazon SQS queue is a temporary repository for messages that are awaiting processing.

The SQS queue acts as a buffer between the component producing and saving data, and the component receiving the data for processing.



This is known as decoupling / loose coupling and helps to enable elasticity for your application.

Amazon SQS is pull-based, not push-based (like Amazon SNS).

Amazon Simple Workflow Service (Amazon SWF)

Amazon Simple Workflow Service (SWF) is a web service that makes it easy to coordinate work across distributed application components.

Amazon SWF is used for processing background jobs that have parallel or sequential steps.

You can think of Amazon SWF as a fully managed state tracker and task coordinator.

Use Amazon SWF if your app's steps take more than 500 milliseconds to complete, you need to track the state of processing, or you need to recover or retry if a task fails.

With SWF you can create distributed asynchronous

systems as workflows.

Tracks the state of your workflow which you interact and update via API.

Best suited for human-enabled workflows like an order fulfilment system or for procedural requests.

AWS recommends that for new applications customers consider AWS Step Functions instead of SWF.

Amazon MQ

Amazon MQ is a managed message broker service for ActiveMQ.

Amazon MQ supports industry-standard APIs and protocols so you can migrate messaging and applications without rewriting code.

Amazon MQ provides cost-efficient and flexible messaging capacity.

Amazon MQ manages the administration and maintenance of ActiveMQ brokers and automatically provisions infrastructure for high availability.



AWS Step Functions

AWS Step Functions can be used to coordinate the components of distributed applications as a series of steps in a visual workflow.

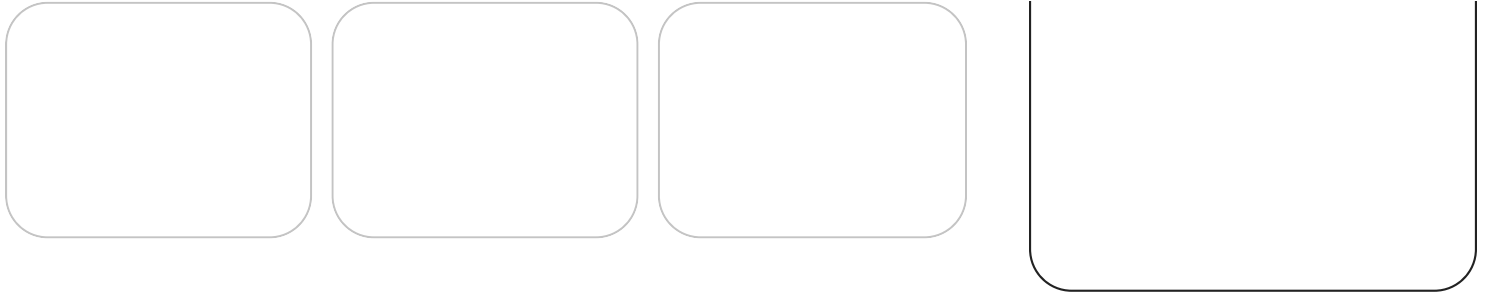
You can quickly build and run state machines to execute the steps of your application in a reliable and scalable fashion.

How it works:

1. Define the steps of your workflow in the JSON-based Amazon States Language. The visual console automatically graphs each step in the order of execution.
2. Start an execution to visualize and verify the steps of your application are operating as intended. The console highlights the real-time status of each step and provides a detailed history of every execution.
3. AWS Step Functions operates and scales the steps of your application and underlying compute for you to help ensure your application executes reliably under increasing demand.

It is a managed workflow and orchestration platform.

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